



CvSU Mission

Cavite State University shall provide excellent, equitable, and relevant educational opportunities in the arts, sciences and technology through quality instruction and responsive research and development activities.
It shall produce professional, skilled and morally upright individuals for global competitiveness.

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DEPARTMENT OF INDUSTRIAL AND INFORMATION TECHNOLOGY ASYNCHRONOUS ACTIVITY 1 ELECTRICAL UNITS

NAME: Bustamante Khier D. _____ DATE: 8-19-26
PROGRAM/YEAR/SECTION: BSIndT 1B _____ SCORE: _____

Objective

After completing this assignment, students should be able to:

1. Identify common electrical quantities and their corresponding units.
2. Define the SI units used in electrical and electronic systems.
3. Distinguish between electrical quantities such as voltage, current, resistance, power, and energy.
4. Apply basic electrical units in simple calculations.

Instructions

Identify the following Electrical Quantities, Symbols, SI Units and Unit Symbols of the following:

Electrical Quantity	Symbol	SI Unit	Unit Symbol
Electric Charge	Q	Coulumb	C
Electric Current	I	Ampere	A
Voltage	V	Volt	V
Resistance	R	Ohm	Ω
Conductance	G	Siemens	S
Power	P	Watt	W

Electrical Energy	E	Joule	J
Capacitance	C	Farad	F
Inductance	L	Henry	H
Frequency	F	Hertz	Hz

Part II – Unit Conversion

Solve the following, then write your answer on the blank before each number:

2,500V 1. Convert 2.5 kV to volts.

0.35A 2. Convert 350 mA to amperes.

4,700,000 Ω 3. Convert 4.7 M Ω to ohms.

2.5 kW 4. Convert 2,500 W to kilowatts.

5 kM 5. Convert 0.005 A to milliamperes.

3,300 Ω 6. Convert 3.3 k Ω to ohms.

0.00075 F 7. Convert 750 μ F to farads.

2 mH 8. Convert 0.002 H to millihenries.

5,000,000 Hz 9. Convert 5 MHz to hertz.

9,000,000 J 10. Convert 2.5 kWh to joules.

Part III – Application

Answer briefly:

1. What is the difference between voltage and current? **-Voltage through a conductor and the current through a conductor.**
2. Why is resistance measured in ohms (Ω)? **It is measured in ohms (Omega) to honor Georg Simon Ohm, the German physicist who discovered Ohm's Law in 1827.**
3. What is the relationship between voltage, current, and resistance according to Ohm's

Law?- **When voltage increases, current also increases (as long as the resistance remains the same) and When resistance increases, current decreases. (as long as the voltage remains the same)**

4. What is the difference between electrical power and electrical energy?-**electrical power consumed over a specific duration of time and electrical energy is converted or consumed per unit of time.**
5. Why is it important for an Industrial Technology student to understand electrical units?-**For safety knowing correct voltage, current, and power ratings prevents electrical overloads, short circuits, component damage, and potential hazards like fires or electric shocks.**